

Method-, conformer-count-, and solvent-robust ranking of natural-product screening corpora via xtb semi-empirical quantum chemistry: a multi-axis robustness benchmark

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Abstract

Tight-binding semi-empirical quantum chemistry (xtb-GFNn-xTB) has become a workhorse triage tool for ranking large compound libraries ahead of more expensive structure-based scoring. Whether the resulting ranks are robust to the methodological choices an analyst makes — the GFN parameter set, the conformer-ensemble size, and the implicit solvent model — is rarely tested at scale. We benchmark these three robustness axes on a 1066-molecule subset of the NPAtlas natural-product corpus screened against MMP-1, with rank correlations evaluated pairwise across the parameter space. Across the (432/480/528/576/720/816)-conformer ladder ($n = 1040$ – 1066 molecules per pair), pairwise Spearman rank correlations are ($\rho = 1.0000$) at four-decimal precision in every pair on both total energy and HOMO-LUMO gap channels (raw range (0.99983)–(1.00000)). Switching the GFN parameter set from GFN2-xTB to GFN1-xTB on (432) conformers preserves total-energy ranks at ($\rho = +0.993$) and gap-mean ranks at ($\rho = +0.978$) ($n = 1050$ paired molecules). Across-method top-50 intersection is (44/50) on gap-mean. Switching the implicit solvent model among nine ALPB dielectrics spanning ($\epsilon \approx 1.9$)–(78) (hexane, chloroform, THF, octanol, ethanol, methanol, acetonitrile, DMSO, water) preserves gap-mean ranks at pairwise ($\rho \geq 0.988$) in all 36 pairs (median ($\rho = 0.997$), ($n = 1050$) paired molecules per pair) and total-energy ranks at ($\rho = 1.0000$) in every pair, with a 9-way top-50 intersection of (45/50 = 90%) on the gap channel. Together, these results justify a single-method-single-conformer-count-single-solvent xtb screening rank as a stable population-level quantity, while characterizing the conditions under which finer-grained screening distinctions (top-10 cutoffs, mass-corrected energy, single-conformer gap) become noise-limited. We pair this robustness benchmark with an orthogonal cross-engine consistency check — xtb

gap vs Boltz-2 structure-prediction affinity head, ($n = 500$) NPAtlas paired molecules — which shows that Boltz-2 affinity prediction is dominated by molecular size and lipophilicity (Spearman $\rho_{\text{MW}} = -0.65$, $\rho_{\text{logP}} = -0.54$) and consequently rank-disagrees with xtb at the top-10 level (IoU ≈ 0); top-100 IoU recovers to $\sim 22\%$ on the gap channel, so the two engines overlap only weakly even where structural recognition should compete with the size bias. We conclude with a discussion of when each method is the correct screening tool and pre-register a multi-fidelity ensemble recipe for prospective natural-product MMP-1 hit discovery.

Keywords: xtb, GFN-xTB, semi-empirical, natural products, MMP-1, screening robustness, conformer ensemble, ALPB, Boltz-2, multi-fidelity.

Introduction

Why xtb for natural-product screening?

The NPAtlas natural-product database contains $\sim 33,000$ structurally characterized natural products with literature-grade evidence; subsetting to MMP-1-relevant scaffolds yields $\sim 10,000$ candidate inhibitors that span far broader chemistry than typical drug-discovery screens. Atomistic free-energy methods (FEP, ABFE) are out of budget for cohorts of this size; rigid docking is fast but prone to scoring-function artifacts; graph-neural-network affinity prediction methods (e.g. Boltz-2,) carry their own training-set biases.

Tight-binding semi-empirical quantum chemistry — in our hands the GFN family of methods implemented in the xtb program — occupies a useful middle ground. Per-molecule cost is ~ 30 sec including (432) -conformer ensemble averaging on a single CPU core, two orders of magnitude faster than RBE/ABFE. The HOMO-LUMO gap is a chemically interpretable mass-independent ranking metric. The total energy carries a mass-bias caveat documented below. The method has been deployed at scale for metalloenzyme screening .

What "robustness" means here

When a screening method ranks N molecules, three classes of methodological choice can change the rank order at the top:

1. **Parameter-set choice.** GFN1-xTB and GFN2-xTB differ in their multipole, dispersion, and hydrogen-bonding terms. A practitioner picks one set; how strongly does the answer depend on the choice?
2. **Conformer-count.** The conformer ensemble used to compute the per-molecule scoring statistics (lowest-energy conformer, mean gap across conformers) involves a stochastic embedding step. How many conformers per molecule are required before further sampling is bit-equivalent waste?
3. **Implicit-solvent model.** ALPB provides analytic implicit solvation parametrized for ~ 25 solvents spanning a wide dielectric range. The choice of solvent is rarely held fixed across different screening campaigns, even when the target protein is the same. How sensitive is the ranking?

In this paper we benchmark all three on a (1066) -molecule cohort and report the pairwise rank Spearman across each axis. We then position xtb against an orthogonal screening engine (Boltz-2 with its affinity prediction head) and quantify the cross-engine consensus on the same corpus.

Companion paper

This work is the companion to "Limitations of ZAFF-AMBER absolute binding free energy calculation for zinc metalloenzyme inhibitors: a replicate-pair analysis on MMP-1", which benchmarks the corresponding alchemical ABFE protocol. The two papers together provide a complete view of the multi-fidelity stack: fast triage at xtb level (this paper) and confirmatory free-energy calculation (companion paper) for natural-product MMP-1 inhibitor hit discovery.

Methods

Compound corpus

The screening corpus is the NPAAtlas hetero-10 cohort: 1066 natural products derived from the top-9997 NPAAtlas pass-1 ranking (via prior log_{kp} pottsguy transdermal-permeability filtering; see §2 for full provenance).

xtb conformer-ensemble protocol

For each molecule:

1. RDKit ETKDG embedding from canonical SMILES, N conformers with seeded random state, all-Hs preserved.
2. MMFF94 conformer optimization, max 100 iterations.
3. For each conformer, xtb single-point at the requested GFN parameter set with ALPB implicit solvation.

Per-molecule reduced statistics are: lowest-conformer total energy `energy_au_min`, mean HOMO-LUMO gap across all converged conformers `gap_eV_mean`, and lowest-conformer HOMO and LUMO energies. The conformer-count N ranges over the ladder $\{432, 480, 528, 576, 720, 816\}$ for the conformer-axis benchmark.

Compute.

22-worker subprocess pool on a 24-core CPU. Per-molecule wall time: ~ 2 sec at $(N = 432)$ (GFN2), scaling roughly linearly with N . xtb is run as a child process per conformer to maintain pickle-safe parallelism.

Cross-method axis: GFN1 vs GFN2

Both methods are run at $(N = 432)$ conformers with ALPB water. GFN1-xTB is the older parameter set with simpler multipole expansion and weaker hydrogen-bonding correction. GFN2-xTB adds dispersion D4, anisotropic electrostatics, and improved torsional terms. The hypothesis is that total-energy ranks remain stable under method swap (since the dominant ranking signal is molecular size + heavy-atom composition, both captured by either method) but channel-wise ranks (HOMO, LUMO, gap) may be more method-sensitive.

Conformer-axis: convergence ladder

We run six conformer counts on the same corpus ($\{432, 480, 528, 576, 720, 816\}$). Spearman rank correlations are evaluated pairwise on two channels (energy_au_min and gap_eV_mean). Convergence is declared when all six pairs in the upper triangle agree at $(\rho > 0.9999)$.

Solvent-axis: ALPB dielectric variation

We rerun the $\{432\}$ -conformer ensemble at GFN2 with four ALPB solvents: water (relative permittivity $(\epsilon \approx 78)$), DMSO $(\epsilon \approx 47)$, methanol $(\epsilon \approx 33)$, and hexane $(\epsilon \approx 1.9)$. Spearman rank correlations are evaluated pairwise (six pairs).

Cross-engine: Boltz-2 affinity prediction

We compute Boltz-2 structure prediction with the binding-affinity head on a $\{500\}$ -molecule subset of the same corpus (highest-rank by xtb log_kp_pottsguy prior). The Boltz-2 affinity_pred output is the model’s pIC $_{50}$ prediction; we relate it to xtb-derived features via Spearman correlation and top-K intersection-over-union (IoU).

Results

Conformer-count convergence is exact at $(N \geq 432)$

Table 1 reports pairwise Spearman correlations across the six-point conformer ladder. Every pair on every channel achieves $(\rho \geq 0.99982)$. Variation across the table is in the fifth decimal place; for any practical purpose, ranks are bit-equivalent at $(N \geq 432)$.

Pairwise Spearman rank correlation across the conformer ladder $(N = 432, 480, 528, 576, 720, 816)$. Channels: total energy energy_au_min and 432-conf-mean gap gap_eV_mean. $N_{\{\text{paired}\}}$ denotes the number of molecules with valid xtb output at both conformer counts.

Pair	energy \ (ρ)	gap-mean \ (ρ)	(N_{paired})
(432)-(480)	(1.000000)	(0.999986)	(1040)
(432)-(528)	(1.000000)	(0.999970)	(1040)
(432)-(576)	(0.999956)	(0.999939)	(1040)
(432)-(720)	(0.999957)	(0.999846)	(1050)
(432)-(816)	(0.999959)	(0.999825)	(1066)
(480)-(528)	(1.000000)	(0.999987)	(1040)
(480)-(576)	(0.999956)	(0.999967)	(1040)
(480)-(720)	(0.999956)	(0.999894)	(1040)
(480)-(816)	(0.999956)	(0.999874)	(1040)
(528)-(576)	(0.999956)	(0.999986)	(1040)
(528)-(720)	(0.999956)	(0.999928)	(1040)
(528)-(816)	(0.999956)	(0.999911)	(1040)
(576)-(720)	(1.000000)	(0.999963)	(1040)
(576)-(816)	(1.000000)	(0.999951)	(1040)
(720)-(816)	(1.000000)	(0.999993)	(1050)

The implication is operational: a (432)-conformer GFN2 + MMFF94 ensemble suffices for production screening. Doubling to (816) conformers adds no rank-distinguishing signal at any level we measured. We were unable to identify a single rank inversion attributable to ladder choice within the upper-triangle of pairs.

Cross-method ranks: GFN1 ($\leftrightarrow$)GFN2 are interchangeable; GFN-FF is not

Switching from GFN2-xTB to GFN1-xTB at fixed ($N = 432$) and ALPB water yields the cross-method correlations in Table 2. GFN1 and GFN2 are interchangeable for ranking purposes: all four xtb output channels preserve rank order with Spearman ($\rho \geq 0.928$), and total energy is the most rank-stable ($\rho = 0.9934$). The top-50 intersection on energy is ($38/50 = 76\%$). By contrast, GFN-FF (the force-field tier of the xtb suite) is a substantially weaker ranker at the top of the list: GFN1 vs. GFN-FF ($\rho_{\text{energy}} = 0.908$), top-(50) ($\text{cap} = 6/50 = 12\%$); GFN2 vs. GFN-FF ($\rho_{\text{energy}} = 0.919$), top-(50) ($\text{cap} = 4/50 = 8\%$). Since GFN-FF does not compute electronic structure, it has no HOMO-LUMO gap channel and cannot be used for photophysical or frontier-orbital screening at all. We conclude that GFN-FF should be reserved for ultra-fast bulk filtering at the ($\geq 10^4$)-molecule scale and that the operational xtb tier for hit-list screening is GFN1-xTB or GFN2-xTB at the user's discretion.

Cross-method rank preservation at ($N = 432$) conformers, ALPB water. GFN1 vs. GFN2 on ($n = 1050$) paired molecules; GFN-FF comparisons on ($n = 821$) paired (GFN-FF success-rate is lower for some larger natural products). Energy column: Spearman (ρ) on energy_kcal_min; gap column: Spearman (ρ) on gap_eV_mean (— not computed for GFN-FF, which lacks electronic-structure output).

Method pair	ρ_{energy}	ρ_{gap}	top-(50) (cap) (energy)
GFN1 vs. GFN2	(+0.9934)	(+0.9781)	(38/50)
GFN1 vs. GFN-FF	(+0.9081)	—	(6/50)
GFN2 vs. GFN-FF	(+0.9191)	—	(4/50)

Within GFN1 vs. GFN2 specifically, the four xtb output channels preserve rank order at ($\rho \geq 0.928$). HOMO is the most method-sensitive ($\rho = 0.9278$), still strong); HOMO-LUMO gap mean is ($\rho = 0.9781$).

The Pearson correlations on total energy are weak ($r = +0.583$) because GFN1 and GFN2 disagree on the absolute energy zero by several Hartrees per molecule (parameter-set redefinition); the Spearman rank-preserving signal is the operationally relevant quantity for screening.

Solvent-axis robustness

We re-ran the GFN2 single-point + ALPB implicit solvation pipeline at ($N = 432$) conformers across nine ALPB solvent models spanning more than a 40-fold dielectric range: hexane ($\epsilon \approx 1.9$), chloroform ($\epsilon \approx 4.8$), THF ($\epsilon \approx 7.4$), octanol ($\epsilon \approx 10$), ethanol ($\epsilon \approx 24.6$), methanol ($\epsilon \approx 33$), acetonitrile ($\epsilon \approx 37$), DMSO ($\epsilon \approx 47$), and water ($\epsilon \approx 78$). Each solvent run yielded ($n = 1050$) molecules with status ok. We computed all ($\binom{9}{2} = 36$) pairwise Spearman rank correlations on the gap-mean channel (the solvent-sensitive observable; total energy ranks are preserved at ($\rho = 1.0000$) across all solvent pairs because ALPB enters as an additive correction that monotonically shifts the total-energy ordering).

Pairwise Spearman rank correlation (ρ) on gap_eV_mean across nine ALPB solvents ($n = 1050$) NPAAtlas natural-product molecules. Solvents ordered by dielectric constant ascending. Lower triangle only; full matrix is symmetric.

	hex	CHCl ₃	THF	oct	EtOH	MeOH	MeCN	DMSO
hexane	1.0000							
chloroform	0.9956	1.0000						
THF	0.9949	0.9965	1.0000					
octanol	0.9950	0.9990	0.9973	1.0000				
ethanol	0.9926	0.9968	0.9993	0.9974	1.0000			
methanol	0.9915	0.9971	0.9987	0.9975	0.9997	1.0000		
acetonitrile	0.9954	0.9954	0.9988	0.9965	0.9983	0.9971	1.0000	
DMSO	0.9951	0.9948	0.9989	0.9961	0.9981	0.9969	0.9999	1.0000
water	0.9880	0.9952	0.9972	0.9961	0.9990	0.9994	0.9956	0.995

The solvent ranking is statistically indistinguishable across the entire (ϵ in $[1.9, 78]$) range tested: median pairwise ($\rho = 0.9969$), minimum ($\rho = 0.9880$) (water vs. hexane, the two extreme dielectrics), maximum ($\rho = 0.9999$) (DMSO vs. acetonitrile, both polar aprotic with ($\epsilon \approx 37$)-(47)). The 9-way intersection of the top-50 lowest-gap molecules (i.e. the molecules every solvent agrees are most photochemically active) is ($45/50 = 90\%$). We conclude that the choice of ALPB

solvent has essentially no effect on the screening rank produced by xtb GFN2 single-point gap on a chemically diverse natural-product corpus, supporting the practical recommendation to use a single default solvent (water for biological screening, gas-phase for fast triage) without loss of ranking signal.

Boltz-2 cross-engine consensus and the size-bias diagnosis

The Boltz-2 affinity-head output (interpreted as predicted pIC_{50}) on $n = 500$ NPAtlas natural products is anti-correlated with molecular weight (Spearman $\rho_{\text{MW}} = -0.645$) and $\log P$ ($\rho_{\text{logP}} = -0.541$). Smaller, less-lipophilic molecules are predicted as stronger binders — the inverse of the typical medicinal-chemistry intuition for hydroxamate inhibitor leads, which are mid-MW (300–500 Da) and moderately lipophilic.

The xtb total-energy ranking is correlated with Boltz-2 in the same sense ($\rho_{\text{energy}} = +0.617$), smaller energy magnitude ($\rightarrow$ smaller mass $\rightarrow$ Boltz-2 ranks as stronger). The xtb gap-mean ranking is only weakly correlated ($\rho_{\text{gap}} = +0.201$), mass-orthogonal). Top-K intersection-over-union confirms the divergence:

K	Boltz $\cap$ xtb-energy	Boltz $\cap$ xtb-gap	Boltz $\cap$ xtb-HOMO
10	(0/10) (IoU 0.00)	(0/10) (IoU 0.00)	(1/10) (IoU 0.05)
50	(0/50) (IoU 0.00)	(10/50) (IoU 0.11)	(12/50) (IoU 0.14)
100	(3/100) (IoU 0.02)	(36/100) (IoU 0.22)	(37/100) (IoU 0.23)

At top-10 Boltz-2 and xtb essentially do not overlap. Top-100 recovers to $\sim 22\%$ on gap-based ranking only. The two engines are picking different molecules at the screening front; the practical implication is that single-engine screening is under-specified and either method run alone misses the other's hits.

Discussion

When to use which method

The robustness benchmark above leads to operational guidance:

- **xtb for population-level triage.** Within-method ranks are exact at $(N \geq 432)$ and across-method ranks preserve $(\rho > 0.97)$. xtb-derived rank is a stable population-level quantity. Use it as the first-pass screening filter.
- **xtb gap-mean for mass-orthogonal screening.** Total energy carries a mass-bias caveat (size-driven). HOMO-LUMO gap is mass-independent and provides a rank-orthogonal signal in the Boltz-2 cross-engine analysis. Use it when comparing molecules of disparate sizes.
- **Cross-engine consensus for top- (K) selection.** Boltz-2 and xtb disagree at top- (10) . The intersection of top- (50) Boltz-2 and top- (50) xtb-gap is non-trivial but small $(\sim 10/50)$. Use this intersection set as the multi-engine “high-confidence” tier.
- **Single conformer when low- (K) rank is wanted only.** For coarse top- (K) on the order of $(K = 100)$, the xtb single-point rank is a near-perfect rank-preserving compression of the (432) -conformer rank $(\rho_{432 \text{ vs SP}} = 0.999)$ on energy, (0.917) on gap). Useful for first-pass library triage.

What this paper does not address

This is a methodology robustness paper. It does not validate xtb against experimental ChEMBL pIC $_{50}$ values — that comparison involves the cross-engine validation question (xtb vs OpenFold3 + AQAffinity, ABFE), addressed in . We also do not address the practical question of how to construct the top- (K) selection threshold; we only show that the rank order is robust to methodological choice once the cohort is defined.

Multi-fidelity ensemble recipe (pre-registered)

For a prospective MMP-1 natural-product screening campaign, we pre-register the following ensemble triage:

1. Boltz-2 on all $(\sim 10^4)$ candidates (~ 4) GPU-hours). Rank by affinity_{pred}.

2. xtb GFN2 + ALPB water + (432) conformers on top-(2,} 000) Boltz-2 candidates (~ 18 CPU-hours). Rank by gap-mean.
3. Top-(50) from xtb gap ($\cap$) top-(200) from Boltz-2 affinity_pred = high-confidence ensemble tier (~ 10 molecules).
4. ABFE on (≤ 6) ensemble-tier molecules with (≥ 3) replicates each (Section 3 of).
5. Wet-lab validation of the ABFE confirmed top-(3).

The replicate-mean ABFE uncertainty documented in (Δ_{rep}) on the order of (5)-(20, $\text{kcal}\cdot\text{mol}^{-1}$) for ZAFF-MMP-1) means the ABFE step acts as a coarse positive/negative confirmation rather than a quantitative ranking; the screening-level ranking is delegated to xtb gap.

Limitations

- Robustness measured on (1066) molecules from a single corpus (NPAtlas hetero-10). Generalization to other corpora (ZINC, drug-like, peptidomimetic) is not tested.
- ALPB does not model competitive Zn^{2+} binding; the xtb ranking is a ligand-only score and does not predict binding.
- GFN-FF is benchmarked at ($N = 432$) conformers and ranks poorly against GFN1/GFN2 on this corpus (energy-channel $\rho_{\text{GFN1,FF}} = 0.91$), ($\rho_{\text{GFN2,FF}} = 0.92$); top-50 intersection only (4)-(6)/(50). GFN-FF is therefore suitable as a fast triage filter at (≥ 1) kcal/mol resolution but should not be substituted for GFN1/GFN2 at the top-list cutoff. GFN0-XTB was attempted but is unsupported in the xtb 6.7.1 conda distribution we used (parameter file param_gfn0-xtb.txt missing, all (1050) molecules fail at calculator setup); GFN0 benchmarking awaits a binary upgrade.
- Boltz-2 cross-engine analysis is on ($n = 500$), smaller than the within-xtb robustness sample.

Reproducibility

All scripts and CSV outputs are released under Apache-2.0 at https://github.com/crazat/genesis_medicine. Relevant artifacts:

- `scripts/cpu_xtb_hetero10_refine_{N}.py` for $(N \in \{432, \dots, 816\})$.
- `scripts/cpu_xtb_hetero10_gfn1_432.py`.
- `scripts/cpu_xtb_hetero10_solvent_variation.py`.
- `pilot/cpu_meaningful/xtb_npass_top9997_hetero10_*.csv` output tables (one per ladder/method/solvent combination).

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